

Lab 10 - Final Project Lab Report

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ESE 3190: Fundamentals of Solid-State Circuits

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Introduction

This report details the theoretical design, simulation, and physical construction of an analog metal detector circuit. The goal of this project was to integrate various fundamental solid-state circuits, specifically LC oscillators, CS and CD amps, and low pass filters into a system that could detect metal objects. The detection mechanism relies on the principle of electromagnetic induction. When a magnetic object enters the electromagnetic field of the variable inductor, eddy currents are induced in the object, which changes the effective inductance of the variable inductor. We can then compare the frequency of the fixed LC oscillator with this variable oscillator to see, or in this case, hear the frequency difference. These two frequencies are then pushed through a mixer stage that takes the sum and difference, and we take the resulting signal and isolate the low frequency signal to drive an audible sound that is driven through a speaker to alert the user of metal.

Design Strategy

The component selection for the resistors, capacitors, and inductors were driven by the target operating frequency of 50kHz for the reference oscillator. To achieve this, I used a 10mH inductor. Using the LC resonance formula, $1/(2\pi\sqrt{LC})$, a 1.01nF capacitor was chosen to pair with a 10mH inductor, establishing a baseline frequency extremely close to the 50kHz target. For the variable inductor, I hand-wound a variable inductor of approximately 7.2 mH in without metal. To match the reference oscillator's frequency, a 1.47nF capacitance was selected for this variable tank, setting its baseline resonant frequency to about 49kHz.

Biasing the active components of the circuit was achieved through a series of tunable potentiometers. The biasing strategy began at the current mirror, where the tail current for the differential pair (summing amplifier) was established. The potentiometer was adjusted to ensure the reference current allowed both input transistors of the differential pair to operate deep within their saturation regions without clipping the incoming AC signals. For the subsequent common-source (CS) amplifiers, the gate biases were tuned to center the quiescent drain voltage between the supply rails, thereby maximizing the allowable voltage swing for the amplified signals. Finally, the common-drain output amplifier was biased to ensure it remained actively conducting while providing sufficient current to drive the low-impedance speaker load.

Hand Calculations

$t_{ox}(cm)$	$9 \cdot 10^{-8}$
W/L	100
$V_{th,n}(V)$	0.7V
$V_{th,p}(V)$	-0.8V
$u_n (cm^2/V \cdot s)$	350
$u_p (cm^2/V \cdot s)$	100
$C_{ox}(F/cm^2)$	$3.9 \cdot 8.854 \cdot 10^{-14} / (9 \cdot 10^{-7}) = 3.837 \cdot 10^{-7}$

$$NMOS (K_n) = \frac{1}{2}(350)(3.837 \cdot 10^{-7})(100) = 6.71 mA/V^2$$

$$PMOS (K_p) = \frac{1}{2}(100)(3.837 \cdot 10^{-7})(100) = 1.92 mA/V^2$$

Frequency

To predict the behavior of the circuit, several theoretical values were derived manually. The frequency of the fixed oscillator can be given by $f_a = 1/(2\pi \cdot \sqrt{LC}) = 1/(2\pi \cdot \sqrt{10 \cdot 10^{-3} \cdot 1.01 \cdot 10^{-9}}) = 50 kHz$. The variable oscillator's frequency was calculated using the ideal resonance frequency equation $f = 1/(2\pi \cdot \sqrt{LC})$. Substituting the measured inductance of 7.2 mH and capacitance of 1.47 nF, the expected frequency ($f = 1/(2\pi \cdot \sqrt{7.2 \cdot 10^{-3} \cdot 1.47 \cdot 10^{-9}}) = 48.92 kHz$). To isolate the resulting beat frequency (the difference between the two oscillators) from the high-frequency carrier frequency, an RC low-pass filter was designed. Using a 100 ohm resistor and a 100 nF capacitor, the filter's cutoff frequency was calculated as $f_c = 1/(2\pi RC) = 1/(2\pi \cdot 10,000 \cdot 1 \cdot 10^{-9}) = 15.9 kHz$. This was the theoretical value I calculated for a low pass filter, but in practice, for the metal detector, I found that removing the caps and not having a low-pass filter allowed the metal detector to work best.

Bias Voltages

The DC operating points across the circuit are configured using five potentiometers (RV1 through RV5). These are important for maintaining the active devices in strict saturation. The summing amplifier (mixer) poses an interesting challenge as it cascades three transistors between

5V supply and ground. To ensure all three devices are in saturation, the current must be restricted. By decreasing the tail current, we ensure that they are in saturation

RV3 Biasing

The RV3 potentiometer sets the gate voltage for the current mirror, establishing the tail current for the differential pair. I targeted a minimal overdrive voltage of about 0.17V to preserve headroom. I calculated the required gate voltage as

$$V_{G3} = V_{th,n} + V_{OV} = 0.7V + 0.17V = 0.87V$$

By tuning the potentiometer to V_{G3} to 0.87V, the resulting tail current is

$$I_{tail} = \frac{1}{2}(350)(3.837 \cdot 10^{-7})(100)(0.87-0.7)^2 = 0.194mA.$$

RV1 Biasing

The RV1 potentiometer controls the DC bias of the differential pair inputs. Assuming perfect symmetry, the 0.194 mA tail current divides equally, resulting in an individual branch current of $I_D = 0.097mA$. To make sure these input transistors remain in saturation, a minimum and maximum gate voltage must be determined.

First, the upper bound ($V_{G1,max}$) is constrained by the drain voltage, which is determined by the voltage drop across the PMOS:

$$0.097mA = 1.92mA/V^2 (V_{SG,load} - 0.8)^2 \rightarrow V_{SG,load} = 1.025V.$$

Since the load is tied to 5V, the drain voltage of the input NMOS is $V_D = 5 - 1.025V = 3.975V$.

Enforcing the saturation condition yields the maximum allowable gate voltage:

$$V_{G1,max} \leq V_D + V_{TH,n} \rightarrow V_{G1,max} \leq 3.975V + 0.7V = 4.675V.$$

Next, the lower bound ($V_{G1,min}$) guarantees that the source node is sufficiently high enough to keep the tail current source in saturation. The tail transistor requires $V_{DS} \geq 0.87V - 0.7V = 0.17V$.

The required gate-to-source voltage for the input pair to pass the branch current is $V_{GS,input} = \sqrt{0.097/6.71} + 0.7 = 0.82V$. Therefore, the minimum gate voltage is $V_{G1} \geq 0.17V + 0.82V = 0.99V$.

Therefore we get a window of $0.99V \leq V_{G1} \leq 4.67V$

RV2 Biasing (First CS Stage)

The first common-source amplifier introduces non-linearity and provides initial voltage gain.

Biased by RV2, it utilizes a PMOS device with a load resistor of $R_D = 2000$ ohms. The saturation requirement is $V_{SD} \geq V_{SG} - |V_{th,p}|$. Given $V_D = I_D R_D$ and a 5V source, this turns into $I_D R_D \leq 5 - V_{G2} - 0.8$.

Substituting the PMOS saturation current equation: $2000 * (0.00192(5 - V_{G2} - 0.8)^2) \leq 5 - V_{G2} - 0.8$.

Dividing out the common $(5 - V_{G2} - 0.8)$ and solving gets us $V_{SG} \leq 1.06V$, which means $V_{G2} \geq 3.94V$. Since the device turns off at $V_G = 4.2V$, the bounding range for this stage is tight:

$$3.94V \leq V_{G2} \leq 4.20V.$$

RV4 Biasing (Second CS Stage)

The second common-source stage is similar but I used a much larger 10kOhmload resistor to maximize voltage gain. Replacing the resistance value in the previous inequality:

$$10000 * (0.00192(5 - V_{G4} - 0.8)^2) \leq 5 - V_{G4} - 0.8$$

This voltage drop creates a smaller biasing window, leading a narrower constraint of $V_{SG} \leq 0.852V$, corresponding to

$$4.15V \leq V_{G4} \leq 4.20V$$

RV5 Biasing (CD Output Stage)

The final driver uses an external IRLB8721 discrete power MOSFET in a CD configuration to buffer the signal into the low-impedance speaker. For this device, the threshold voltage is about $V_{th} = 1.8V$. To maintain saturation, $V_{D} \geq V_G - 1.8V$. Because the drain is connected to the 5V rail, the theoretical bound is 6.8V, but the supply rail is 5V.

To determine the lower bound, we use the saturation equation $I_D = K_n(V_G - V_S - V_{th})^2$ and Ohm's law at the source $V_S = I_D R_L$, we can substitute I_D to get a relationship between the gate and the source voltages:

$$V_S = R_L * K_n (V_G - V_S - 1.8)^2$$

Rearranging this expression to solve for the gate voltage (V_G) yields:

$$V_G = \sqrt{V_S / (R_L * K_n)} + V_S + 1.8.$$

From the equation above, it is evident that as V_s increases, V_g must also increase. Therefore the absolute minimum for V_g occurs when the output voltage approaches 0. Substituting $V_s = 0V$ into the equation gives 1.8V.

Therefore, our biasing is:

$$1.8V \leq V_{G5} \leq 5.0V.$$

Gain stages

The differential gain of the mixer can be approximated by the ratio of the transconductances, $A_v = -g_{m,in}/g_{m,load}$. The current running through both the load transistor and the input transistor is the same, so the differential gain is basically $\sqrt{u_n/u_p} = -1.87$.

For the common-source stages, the small signal gain is defined as $A_v = -g_m R_D$. Selecting a bias point of $V_{G2} = 4.0V$ for the first CS stage yields a source to gate voltage of 1.0V, generating a drain current of 0.0768mA and a transconductance of 0.768mS. The theoretical gain is therefore:

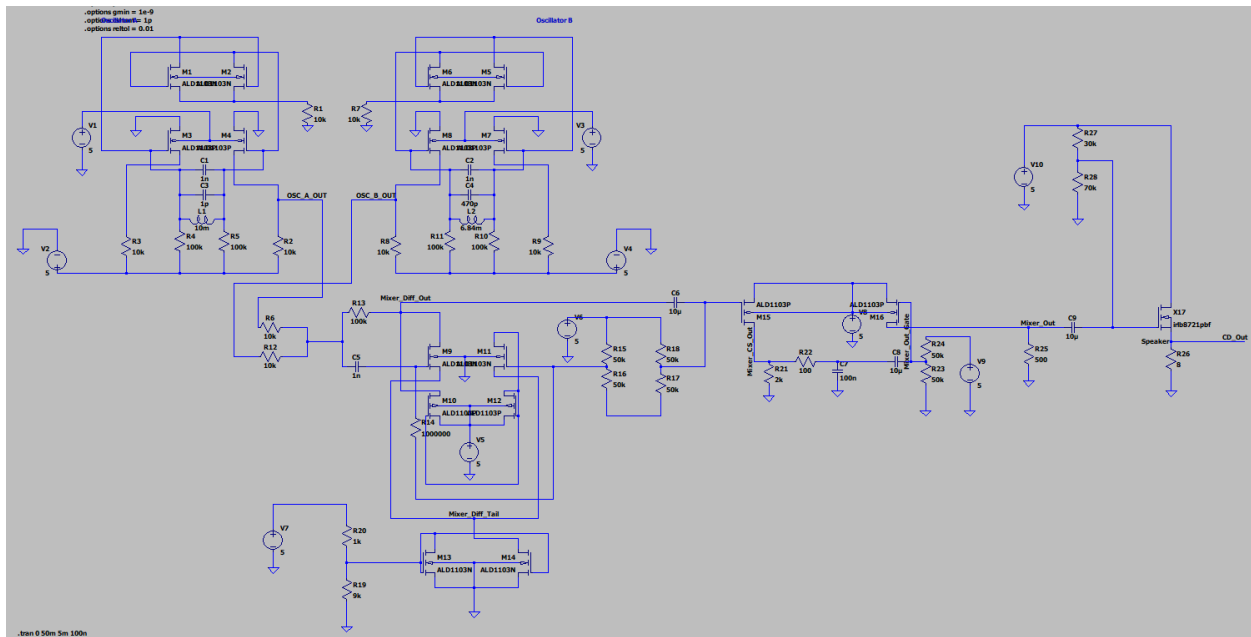
$$A_{v,CS1} = -(0.768mS)(2000 \text{ ohms}) = -1.54V/V.$$

Applying the same approach to the second CS stage, we select a bias of $V_{G4} = 4.17V$ generates a much smaller current of 0.0017mA and a transconductance of 0.115mS, resulting in

$$A_{v,CS2} = -(0.115mS)(10000 \text{ ohms}) = -1.15V/V.$$

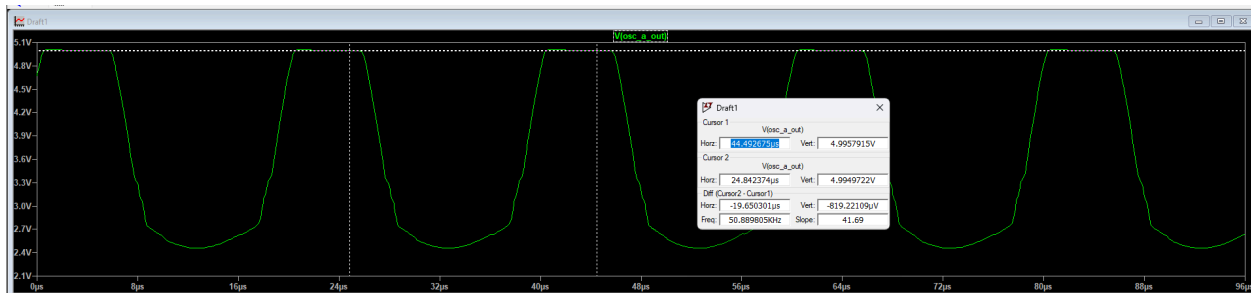
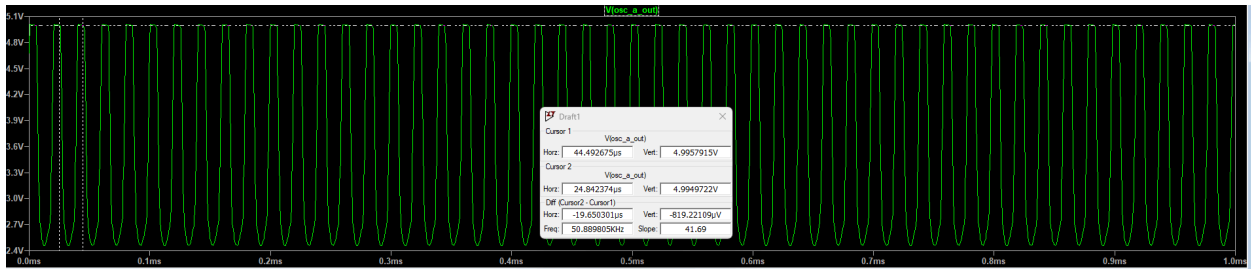
For the CD stage, the exact gain is governed by $A_v = g_m R_L / (1 + g_m R_L)$. Because R_L is only 8 ohms, the gain of this CD amp falls very low below 1V/V.

Simulation Results

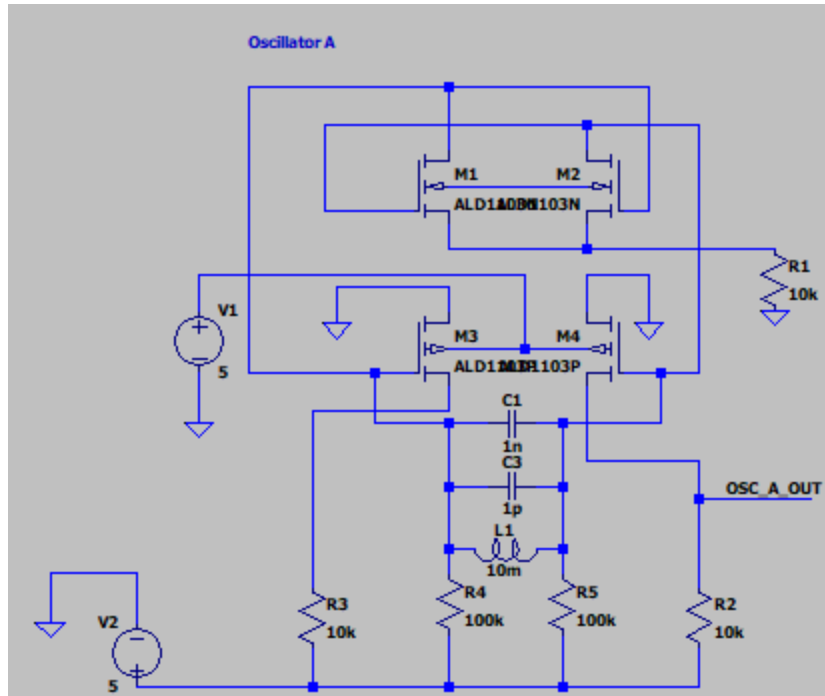


Here is the full LT Spice schematic.

Fixed Oscillator A Oscillation

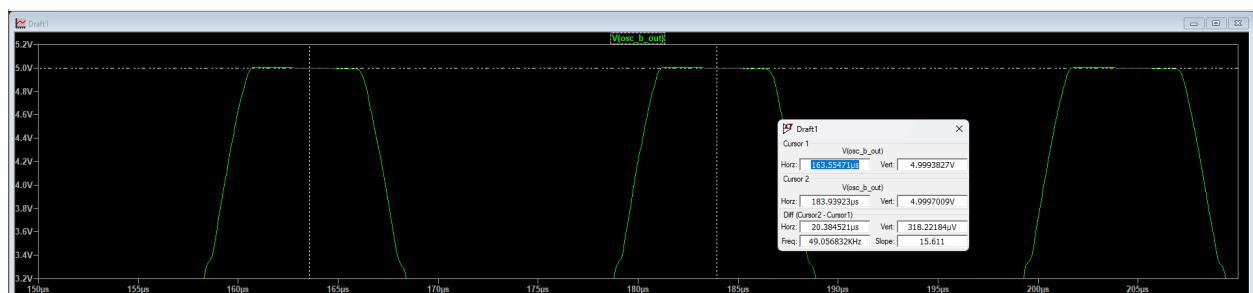
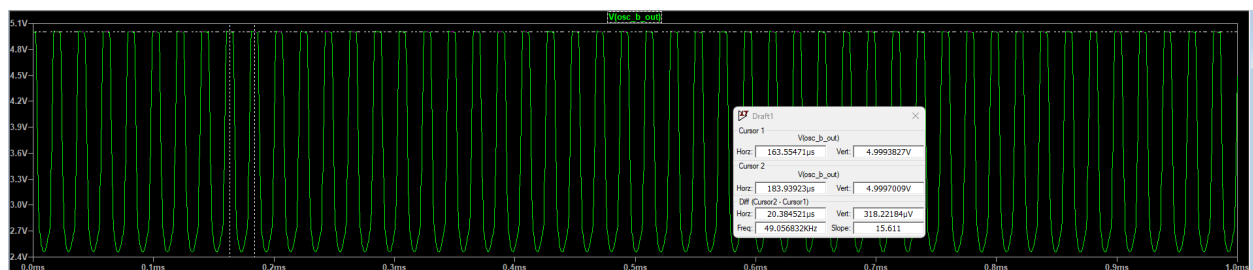


Frequency: 50.889kHz

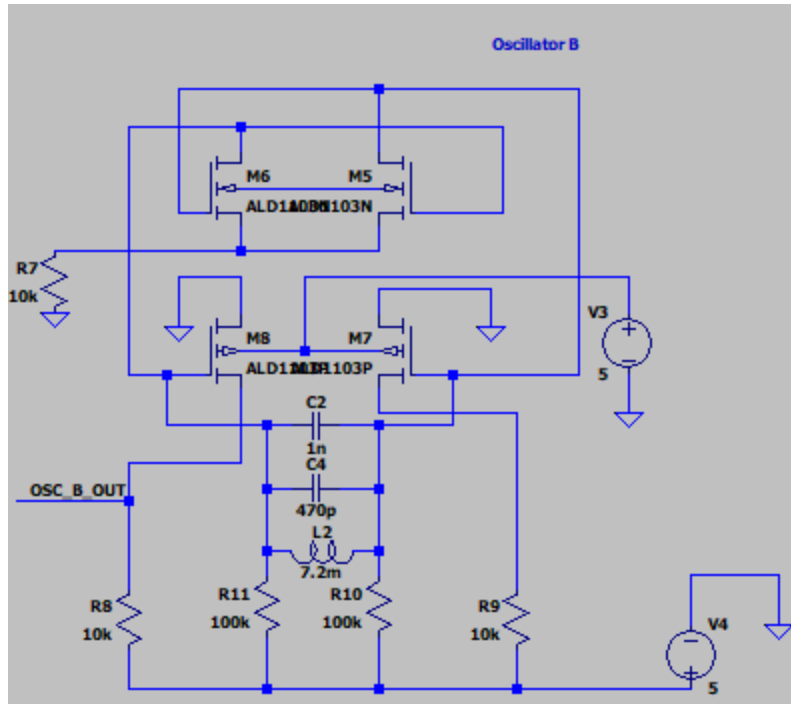


The fixed oscillator, labeled Oscillator A, was designed using a 10mH inductor and oscillated at a measured frequency of 50.889kHz. This value is close to the intended 50kHz and serves as the reference frequency for the rest of the metal detector circuit. We chose 10mH and 1.01nF capacitance to target $1/(2\pi \cdot \sqrt{10 \cdot 10^{-3} \cdot 1.01 \cdot 10^{-9}}) = 50,079.4377979314 \text{ Hz}$, or around 50kHz.

Variable Oscillator B Oscillation

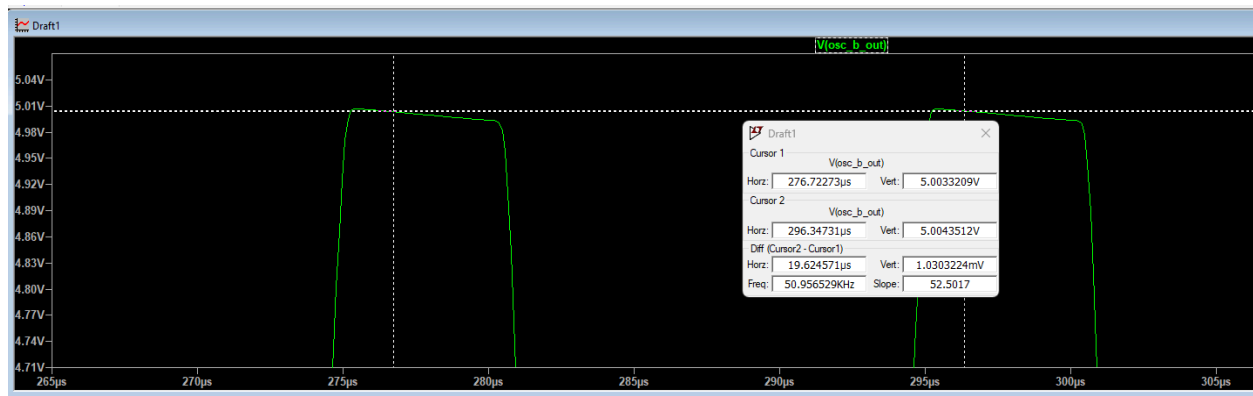


Frequency: 49.0568kHz

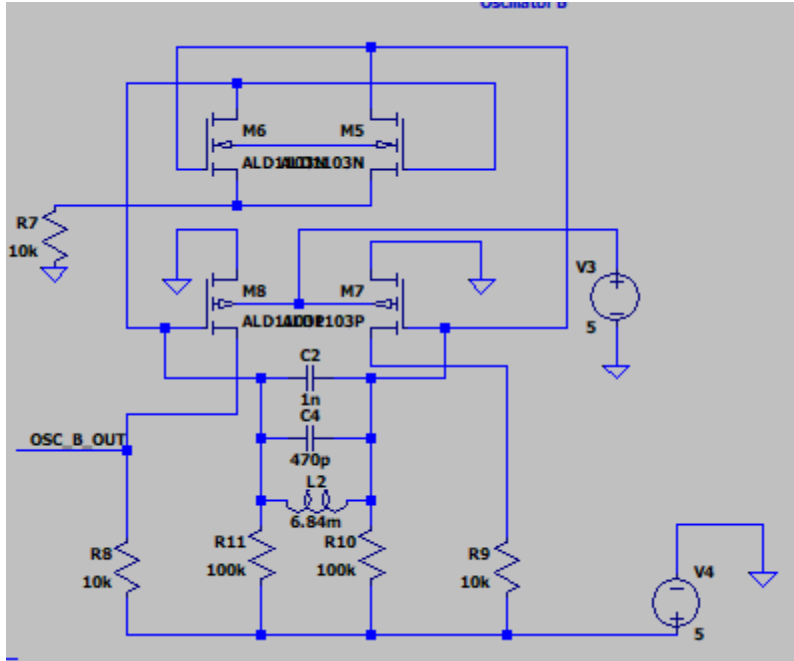


The variable oscillator, labeled Oscillator B, was designed using a 7.2mH inductor and oscillated at 49.0568kHz under normal conditions. This value is close to the intended 49kHz target. We chose 7.2mH and 1.47nF to get about $1/(2\pi \cdot \sqrt{7.2 \cdot 10^{-3} \cdot 1.47 \cdot 10^{-9}}) = 48,920.9876239286$ Hz or around 49kHz.

Variable Oscillator B Oscillation - Inductor Decreases by 5%



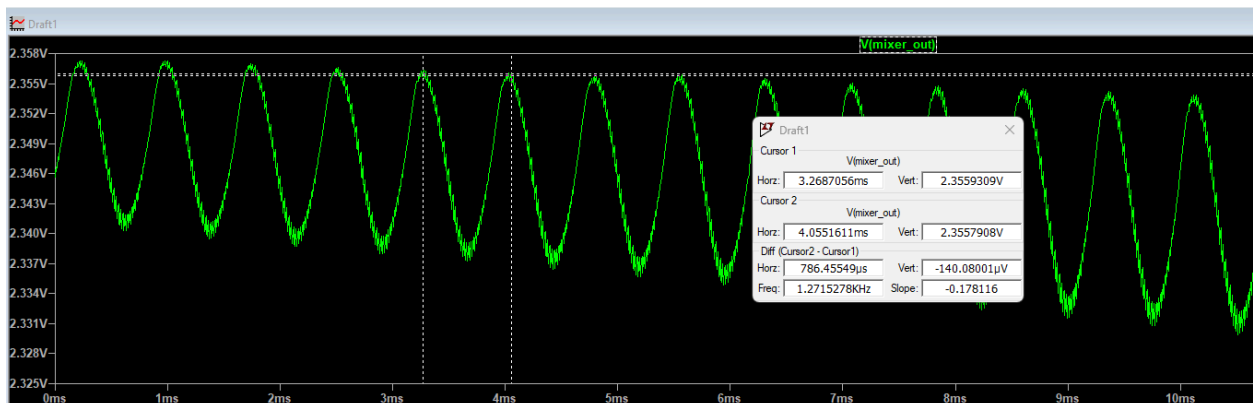
Frequency: 50.9565kHz



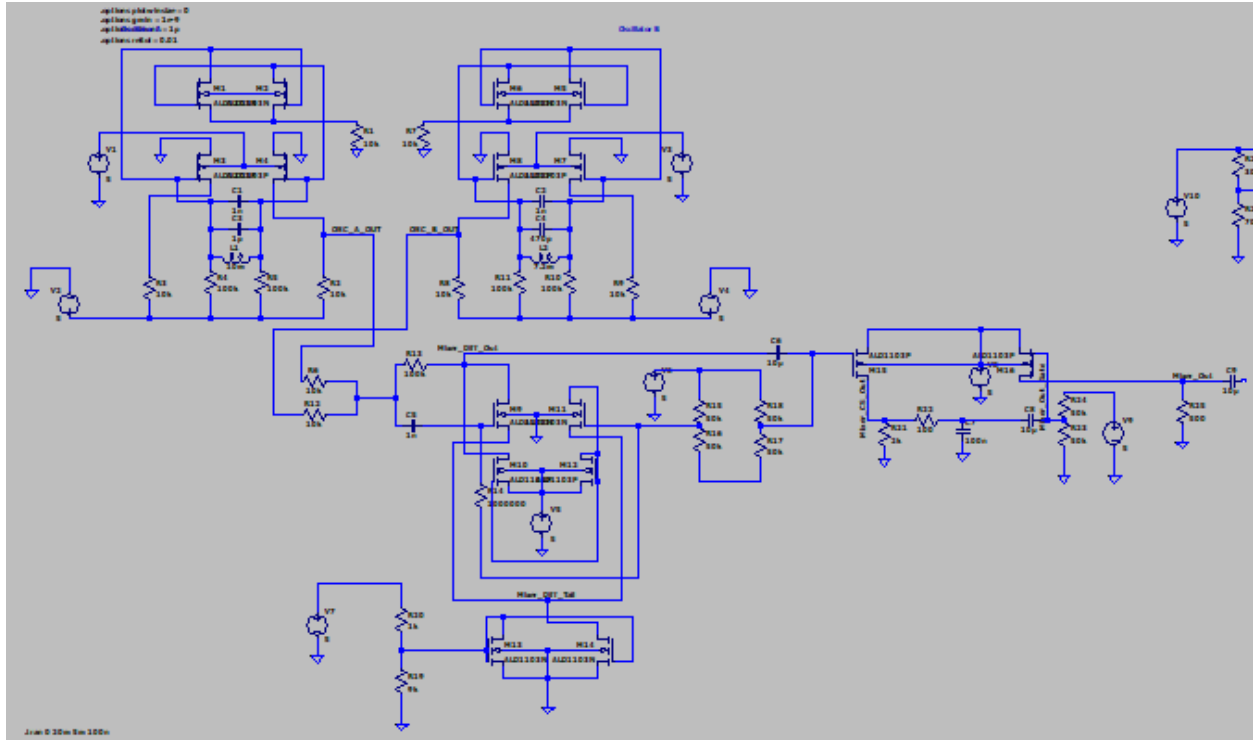
Inductance value went from 7.2mH to 6.84mH.

To model the near-metal condition, the variable oscillator inductance was decreased by 5%, from 7.2mH to 6.84mH. Under this condition, the variable oscillator frequency increased to 50.9565kHz. This makes sense as $1/(2\pi \cdot \sqrt{6.84 \cdot 10^{-3} \cdot 1.47 \cdot 10^{-9}}) = 50,191.8742647764 \text{ Hz}$, or around 50kHz. In this case, the difference between the fixed oscillator and the variable oscillator became much smaller.

Mixer Output

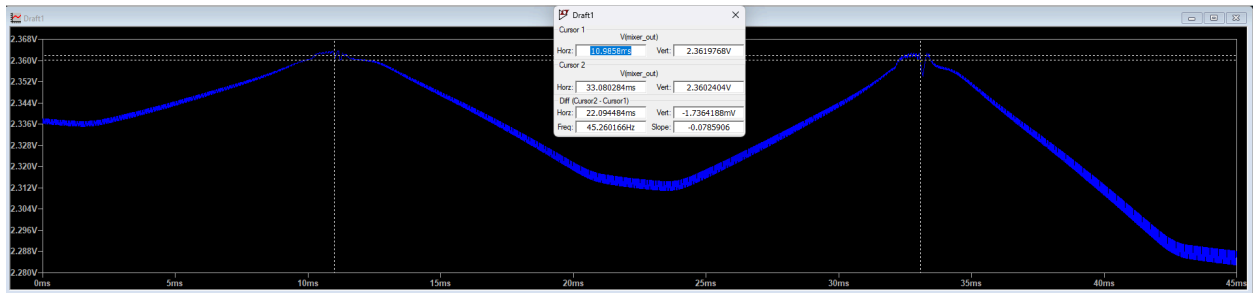


Frequency: 1.27152kHz

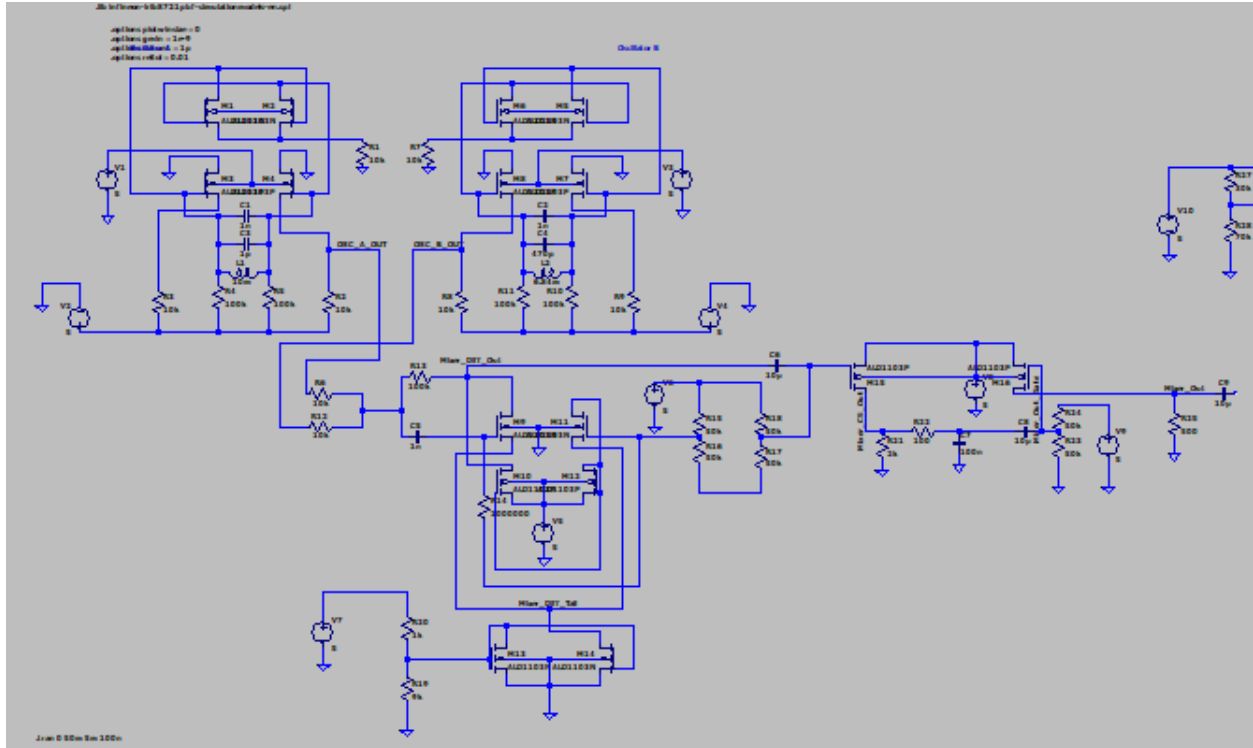


The mixer output was then simulated for both conditions. In the normal case, the mixer output frequency was measured as 1.27152kHz.

Mixer Output-Inductor decreased by 5%.

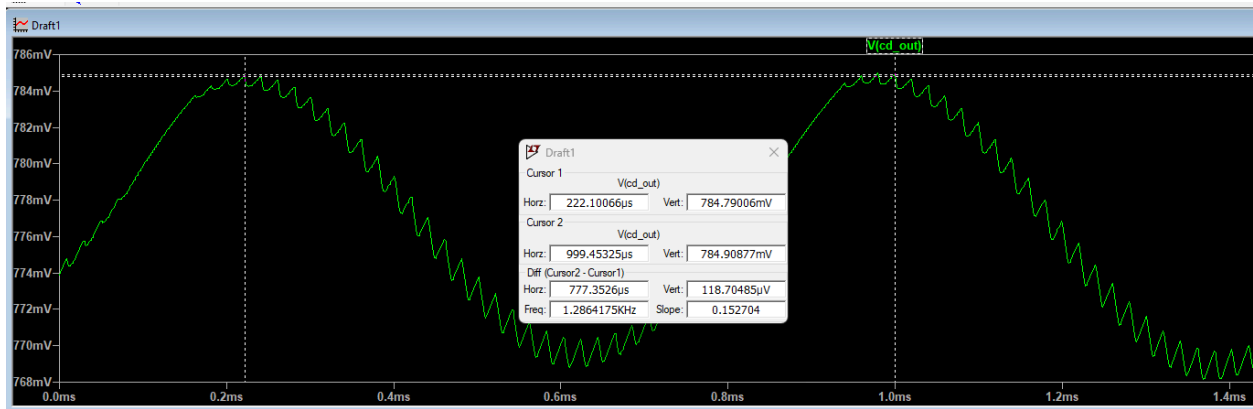


Frequency: 45.260Hz

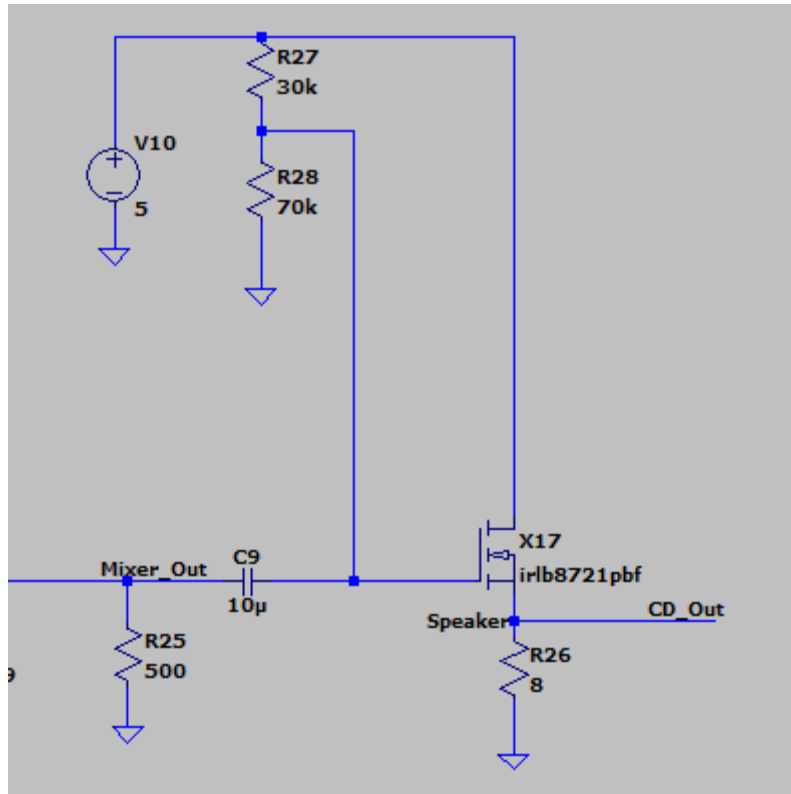


In the reduced-inductance case, the mixer output frequency was measured as 45.260 Hz. These results show that the mixer and filtering stages were able to successfully translate the oscillator-frequency separation into a much lower-frequency signal. Although the measured mixer frequencies are not exactly equal to the direct subtraction of the oscillator frequencies, they are in the expected range and demonstrate the intended behavior of the detector.

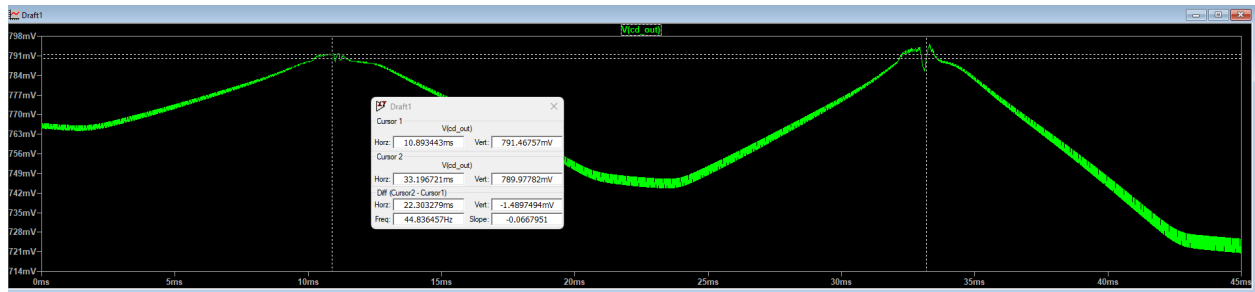
Final Output Stage -Normal



Frequency 1.2864175 kHz



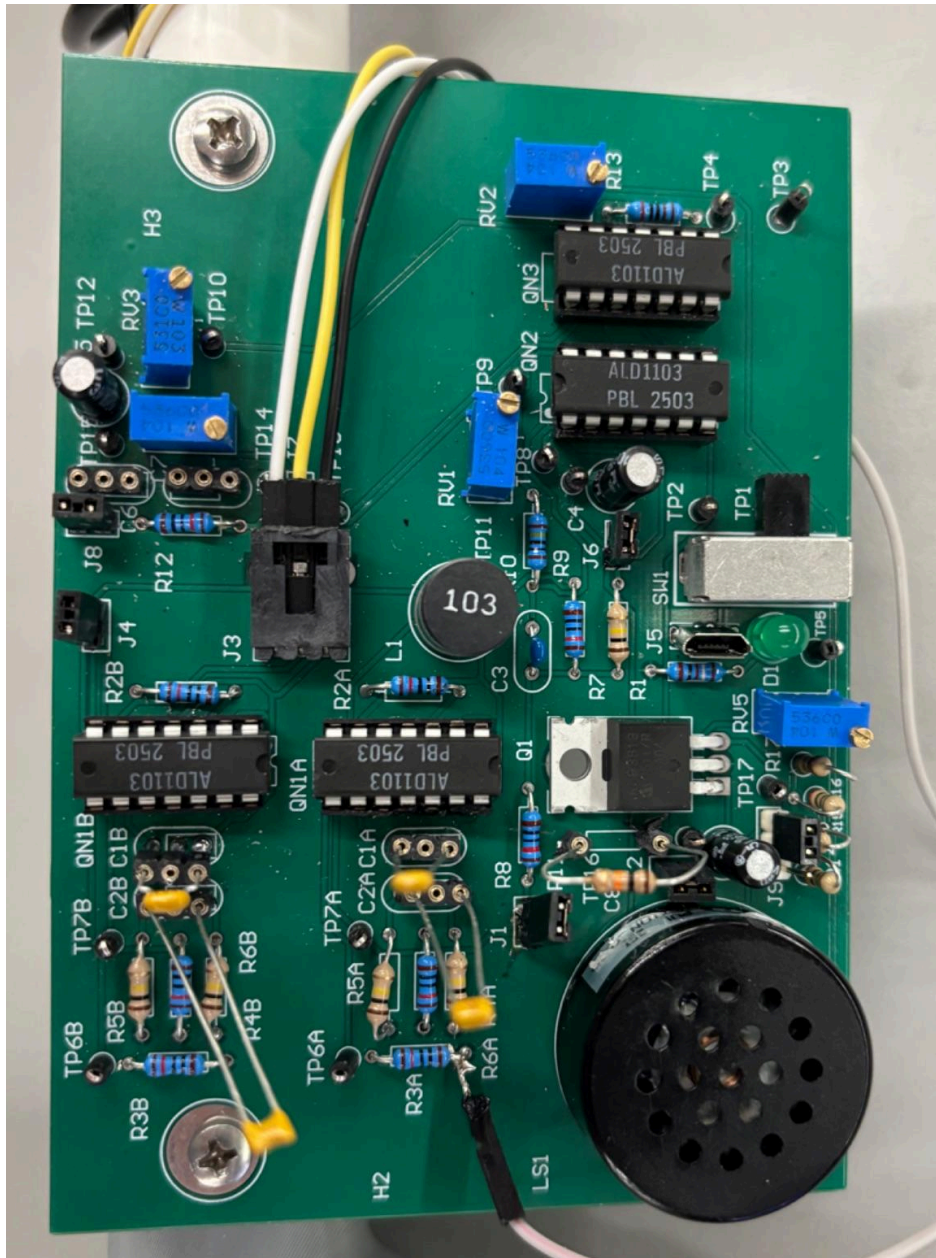
Final Output Stage -Inductor dropped by 5%

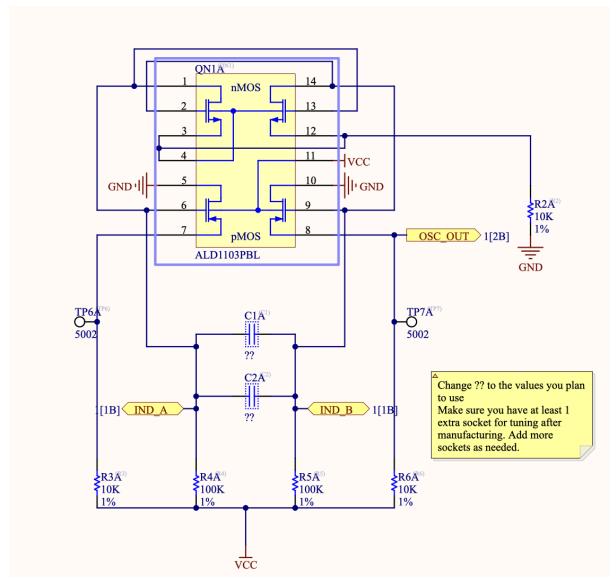


Frequency 44.836Hz

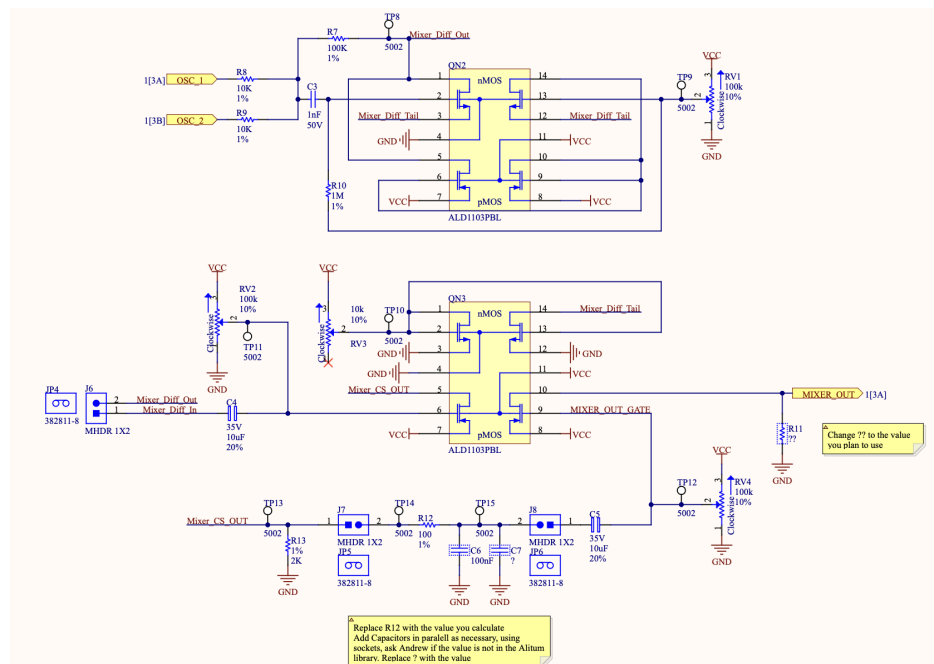
Finally, the output stage was simulated. In the normal case, the final output-stage frequency was 1.2864175kHz. In the reduced inductance case, the final output-stage frequency was 44.836 Hz. These results closely match the mixer-stage outputs, indicating that the later gain and buffer stages preserve the low-frequency information generated by the mixer.

PCB Layout & Schematic





The fixed and variable oscillators follow the same schematic. The fixed oscillator has a built-in inductor while the variable oscillator is connected to a manually coiled inductor. C1A/B and C2A/B values are chosen to get the specific oscillation frequencies as calculated before.



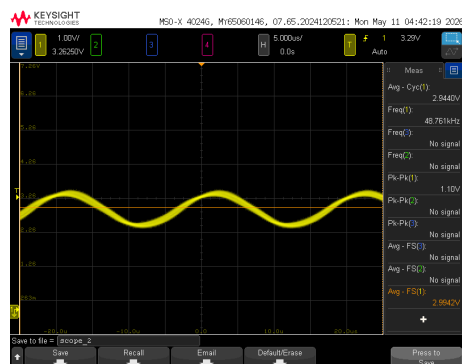
This part of the schematic shows the mixer and CS stage with potentiometers to change the biasing of the various transistors. R11 and C7 values are chosen based off the wanted gain and cut-off frequency respectively.

Oscillator Measurements



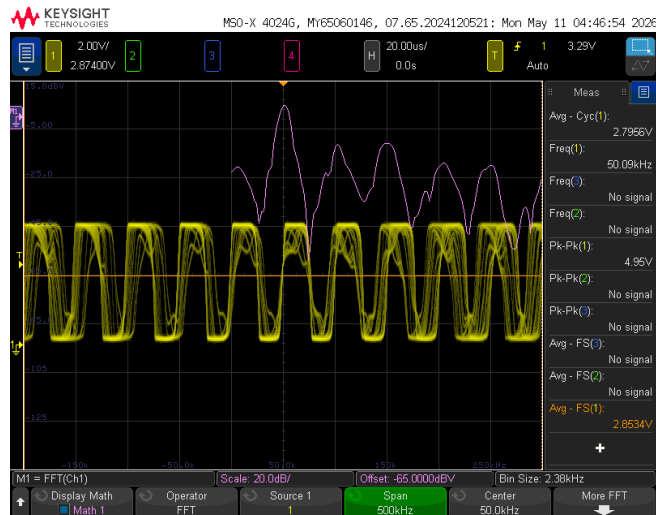
The fixed reference oscillator (left) produced a stable sinusoidal waveform with a measured frequency of 51.106kHz and a peak-to-peak amplitude of 2.34V. The variable oscillator was then measured without any metal present (right) yielding a baseline frequency of 46.534kHz and a peak to peak amplitude of 1.54V.

When a metallic object was introduced to the magnetic field of the hand-wound coil (below), eddy currents induced in the metal created an opposing magnetic field, effectively reducing the coil's inductance. Consequently, the variable oscillator's frequency shifted upwards to 48.761kHz, while its amplitude decreased slightly to 1.10V.

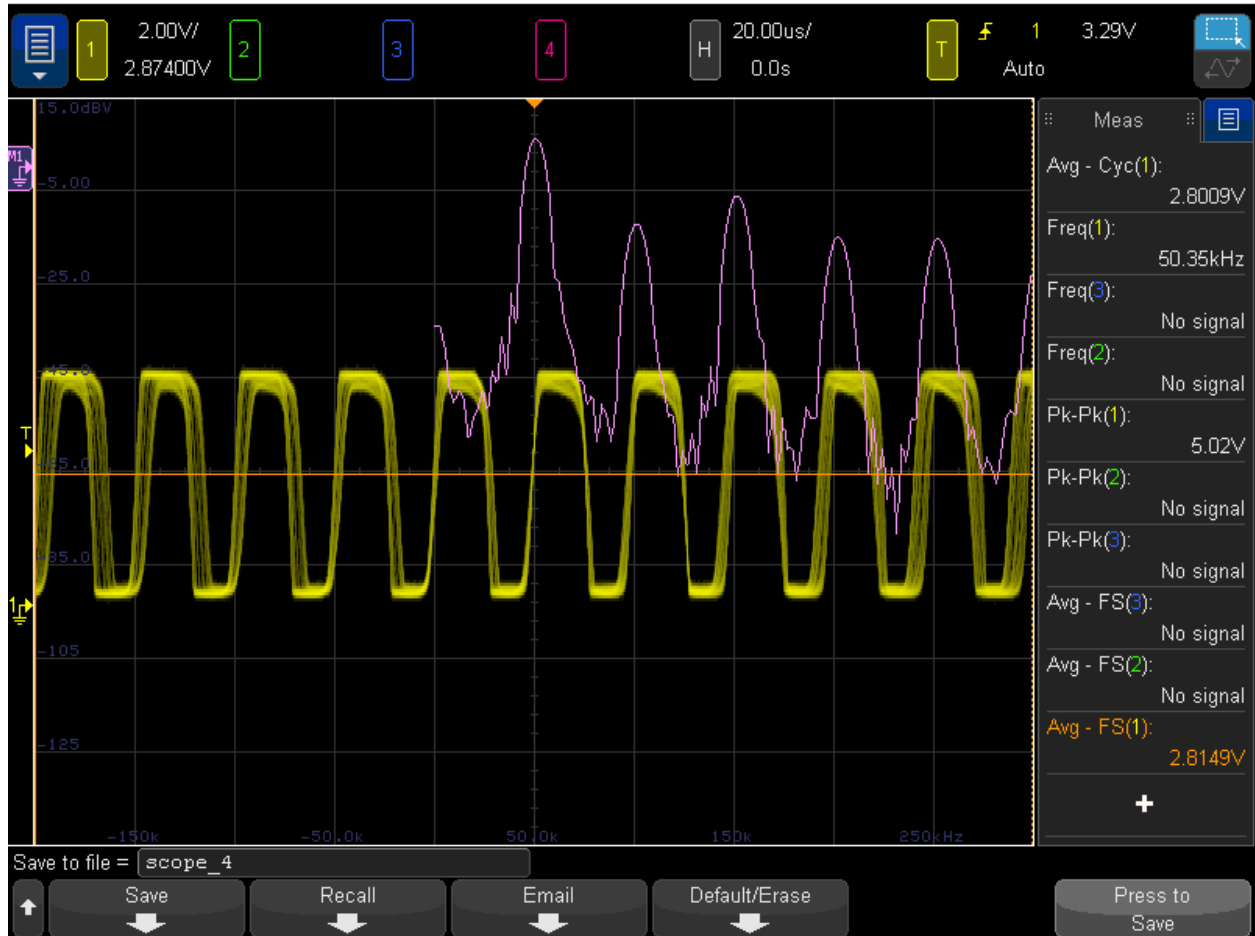


Mixer and 1st CS Amplifier Measurements

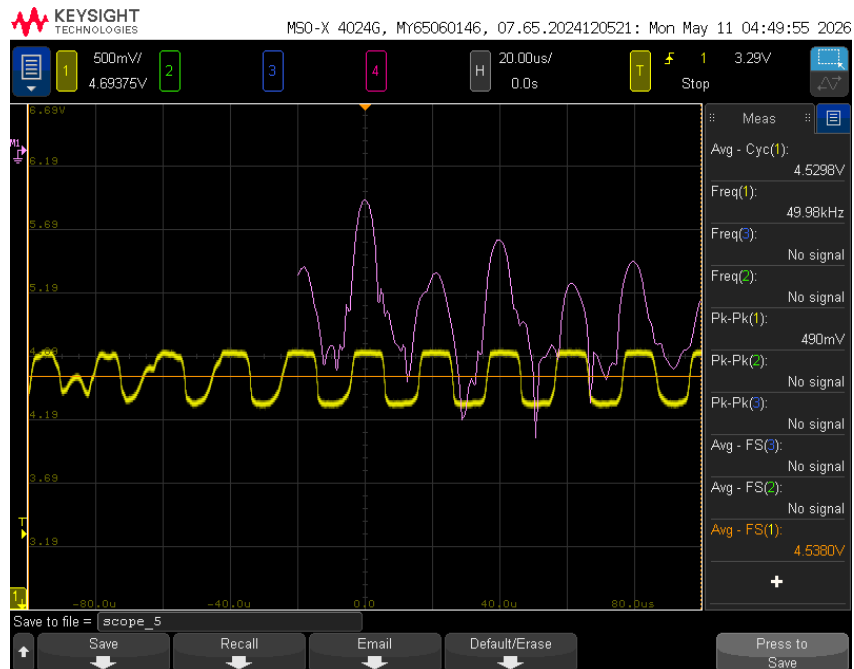
The outputs of the two oscillators were fed into the differential summing amplifier. In the absence of metal (below) the time-domain waveform exhibited a heavily modulated envelope with a peak-to-peak voltage of 4.95V. The FFT trace showed a single dominant peak centered around 50.09kHz, representing the average of the carrier frequency of the two mixed signals.



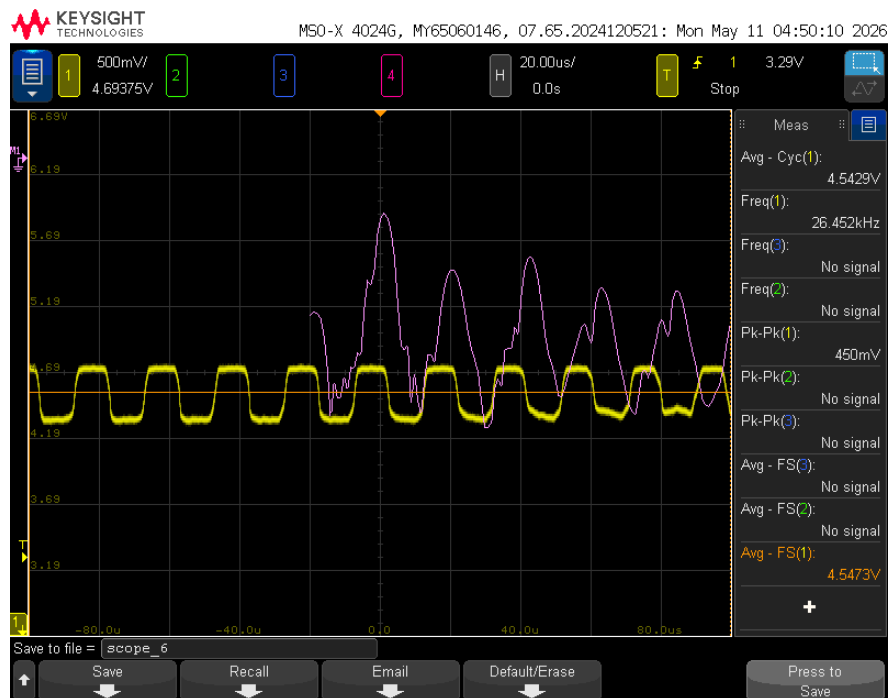
When the metal was introduced, this main carrier peak shifted slightly to 50.35kHz, and the peak-to-peak amplitude of the time-domain signal increased marginally to 5.02V.



The signal then passed through the first common-source (CS) amplifier, which was designed to exploit the MOSFET's square-law behavior to introduce non-linearity necessary for multiplication. Without metal (below), the 1st CS Amp output measured 490mV peak-to-peak, with the FFT still showing a dominant high-frequency peak at roughly 50kHz.

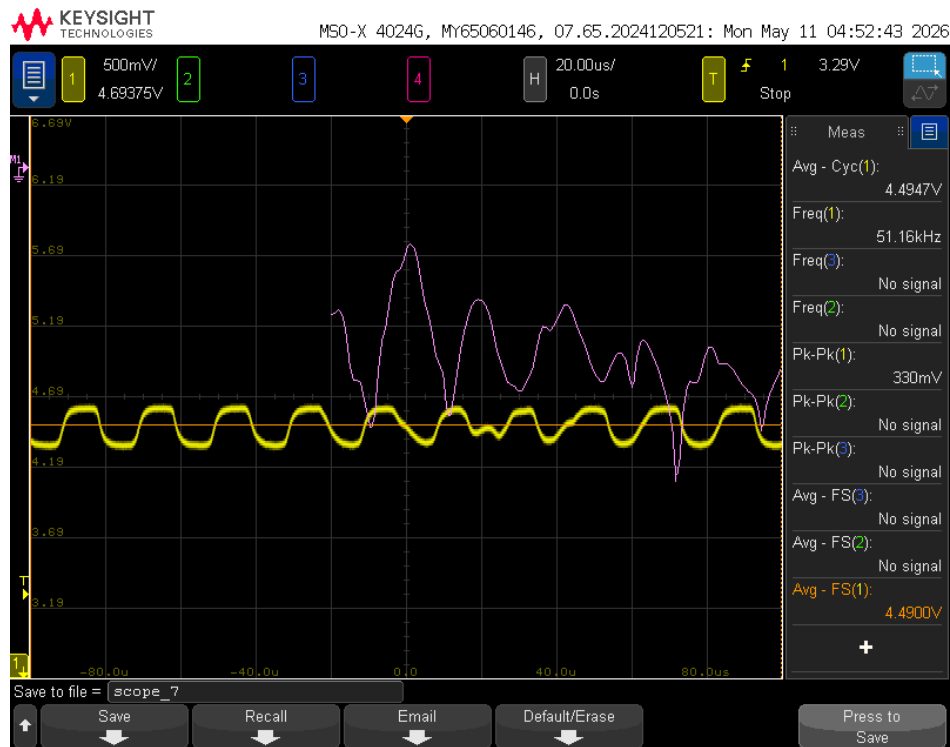


With metal present (below), the amplitude measured 450mV peak-to-peak.

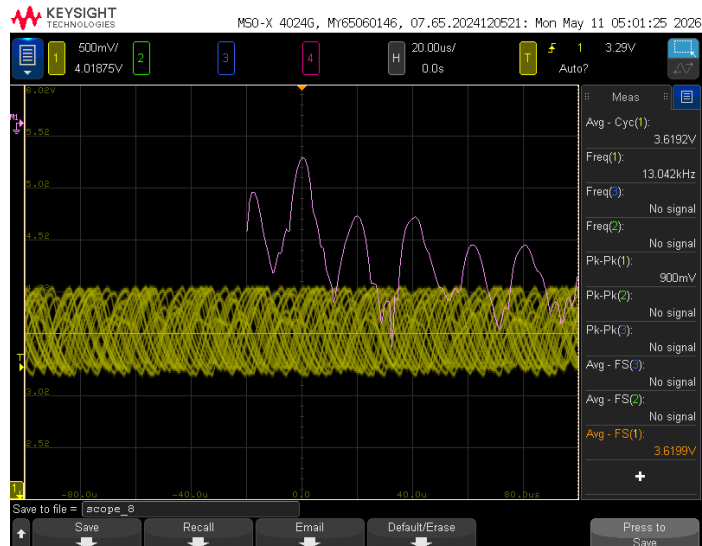


Filter, 2nd CS Amp, and Output Stage Measurements

I found in testing that the traditional RC low-pass filter attenuated the signal too severely to adequately drive the final speaker stage. As a result, I removed the capacitors (below).



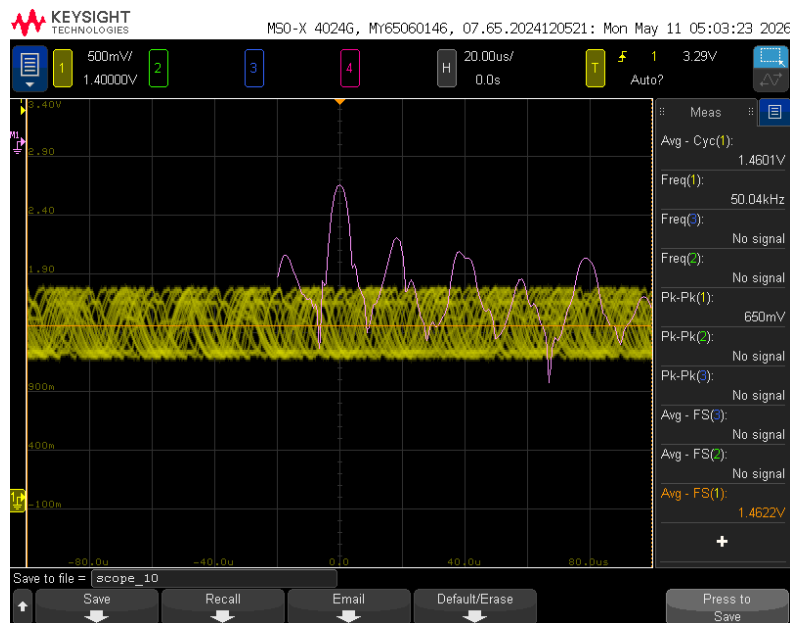
The signal was then boosted by the 2nd CS Amplifier. In the idle state (below), the output was a high-frequency dominant wave measuring 900mV peak to peak.



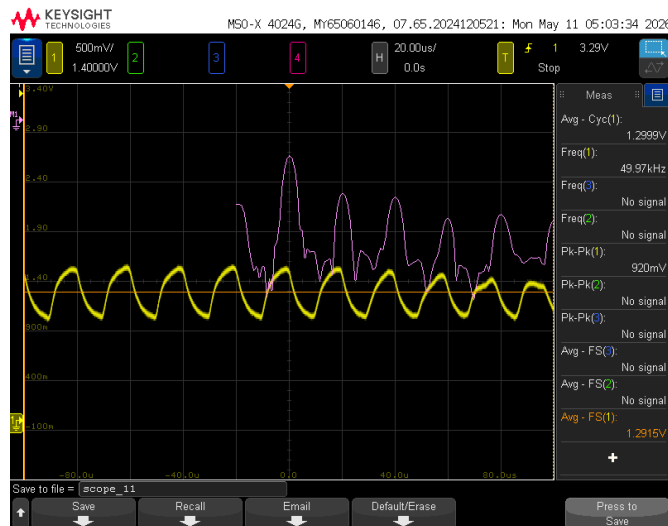
Upon detecting the metal, the waveform maintained a similar amplitude.



Finally, the signal was fed into the common-drain (CD) output buffer. Without metal (below), the CD stage outputs a baseline high-frequency signal at 650mV peak to peak, which is largely inaudible.



However, when metal was introduced (below), the peak to peak voltage increased to 920mV, and we can clearly see a stronger peak in the FFT in lower frequencies.



Experimental vs Simulated Results

The most foundational comparison lies in the oscillator frequencies. In LTSpice, the fixed reference oscillator was simulated at 50.889 kHz, while the physical circuit measured an incredibly close 51.106 kHz, yielding an error of just 0.4%. However, the variable oscillator showed a wider discrepancy; the simulation targeted a frequency of 49.056 kHz, but the physical hand-wound coil produced an idle frequency of 46.534 kHz (Scope_1), an error of approximately 5.1%. This variance is entirely expected, as the physical hand-wound inductor possesses unpredictable parasitic series resistance and capacitance that SPICE models do not capture. Despite this baseline offset, the frequency shift during metal detection was remarkably consistent. In LTSpice, simulating a 5% drop in inductance shifted the variable frequency by +1.9 kHz (from 49.05 kHz to 50.95 kHz). In the physical experiment, introducing metal shifted the frequency by +2.227 kHz (from 46.534 kHz to 48.761 kHz).

Significant deviations between simulation and reality became more apparent in the voltage amplitudes and gain stages of the signal chain. In LTSpice, adjusting the current mirror to keep the differential summing amplifier in saturation is a precise operation. In the physical circuit, the mixer output produced a massive 4.95V peak-to-peak signal, effectively utilizing the entire 5V supply headroom. However, this large swing pushed the transistors to the very edge of the triode region, resulting in slight squaring of the waveform that was not as prominent in the idealized simulation. Furthermore, when this signal was fed into the first common-source (CS) amplifier, the physical output amplitude plummeted to 490mV peak-to-peak, and later to 450mV during detection. This aggressive attenuation highlights the difficulty of

physically biasing a high-gain CS stage; real-world channel-length modulation r_o and component mismatches severely degraded the expected theoretical voltage gain, forcing the transistor to operate with less than ideal transconductance g_m .

The most drastic deviation from the simulation was the removal of the RC low-pass filter. In LTSpice, placing a 100nF capacitor perfectly stripped away the 50 kHz carrier wave, leaving a clean 1–2 kHz beat frequency. In physical testing, introducing these capacitors attenuated the entire signal—including the desired low-frequency beat—too severely to drive the final stage. Finally, the common-drain (CD) output stage exhibited the expected loading effects. While a simulated, unloaded source follower provides a voltage gain near 1 V/V, the physical circuit driving an 8-ohm speaker experienced a massive voltage divider effect due to the MOSFET's internal output resistance ($1/g_m$). As a result, the final output generated 650mV peak-to-peak at idle and 920mV peak-to-peak during metal detection. Despite these substantial drops in voltage amplitude compared to the SPICE environment, the cascade of amplifiers successfully preserved the low-frequency intermodulation envelope generated by the mixer, validating the system.

Discussion

The empirical results of this project validate the core principles of solid-state circuit design, while demonstrating the practical realities and non-ideal behaviors in analog. The heart of the detector, the differential summing amplifier functioning as a frequency mixer, worked well, exploiting the non-linear square-law characteristic of the common-source amplifier to generate the sum and difference frequencies of the two LC oscillators. The slight shift in the variable oscillator's idle frequency compared to the theoretical calculation (46.5 kHz physical versus 49 kHz theoretical) shows the results of parasitic elements; the hand-wound coil introduced series resistance and inter-winding capacitance that the math did not account for. The relative frequency shift upon detecting metal proved to be robust, fulfilling the primary detection criteria.

The most important differences between theory and practice occurred in the amplification and filtering stages. The necessity to remove the RC low-pass filter highlights a design consideration: the trade-off between frequency filtering and maintaining sufficient signal amplitude. By relying on the natural parasitic low-pass characteristics of the cascaded amplifier stages, the system was able to effectively isolate the low-frequency beat without the voltage attenuation introduced by the filter.

Furthermore, the significant gain reductions observed in both the common-source and common-drain stages emphasize the limitations of calculations that assume ideal transistor behavior. The physical circuit

was subjected to channel-length modulation (r_o) and loading effects, particularly at the CD output stage driving the 8-ohm speaker. The voltage divider formed between the output transistor's internal resistance ($1/g_m$) and the speaker load reduced the peak-to-peak voltage swing. However, this behavior perfectly illustrates the intended function of a source follower: while it failed to provide voltage gain, it successfully provided the massive current gain required to drive a heavy, low-impedance speaker without collapsing the previous high-impedance amplification stages.

Conclusion

In conclusion, the design, simulation, and physical construction of the analog metal detector was a success. The project successfully integrated multiple solid-state topologies, including LC oscillator, differential amplification, non-linear frequency mixing, and impedance buffering, into a single, functional metal detection system. While physical constraints, component tolerances, and parasitic loading effects caused the lab voltage amplification to differ from the idealized SPICE simulations, the underlying principles proved to be true. This process provided hands-on experience in bridging the gap between mathematics, idealized software modeling, and the physical realities of analog circuit debugging and implementation.

